

THOMSON, GA, USA

WMO: 720289

Lat: 33.530N

Lon: 82.516W

Elev: 498

StdP: 14.43

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63836

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.8	26.8	3.7	6.8	37.4	9.7	9.2	38.1	19.5	47.6	17.5	46.9	1.5	300	0.359

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	96.8	74.6	94.2	74.6	92.0	74.7	80.2	88.7	79.0	87.4	78.0	86.4	5.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.4	150.3	83.6	77.0	143.2	82.6	75.7	136.7	81.4	44.1	89.1	42.9	87.0	41.7	86.0	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.9	11.2	16.4	98.9	4.1	3.4	13.5	101.3	11.1	103.3	8.8	105.1	5.9	107.6
			15.9	81.1	4.2	2.9	12.9	83.2	10.5	84.9	8.2	86.5	5.1	88.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.9	45.7	49.6	55.7	63.5	71.1	78.4	80.8	79.7	75.0	64.3	52.9	48.6	(6)
	DBStd	14.51	9.93	9.00	9.05	7.60	6.17	4.61	3.72	3.97	5.20	7.79	8.09	9.13	(7)
	HDD50	569	206	111	46	5	0	0	0	0	0	3	58	140	(8)
	HDD65	2483	602	437	313	117	21	0	0	0	3	108	370	512	(9)
	CDD50	5624	71	98	222	409	655	852	954	921	751	447	146	98	(10)
	CDD65	2064	3	4	23	71	211	402	489	456	304	87	9	5	(11)
	CDH74	19977	8	31	241	781	2085	4086	4989	4226	2691	758	63	18	(12)
	CDH80	8686	0	1	41	197	804	1969	2424	1912	1142	192	4	0	(13)
(14) Wind	WSAvg	3.8	4.4	4.7	4.8	4.7	4.0	3.4	3.1	2.9	3.1	3.4	3.5	3.9	(14)
(15) Precipitation	PrecAvg	45.1	4.4	4.1	5.0	2.7	3.1	4.2	4.3	4.4	3.7	3.1	2.9	3.3	(15)
	PrecMax	55.7	7.9	6.9	9.4	7.0	8.7	7.9	8.6	7.5	11.3	10.0	8.7	6.6	(16)
	PrecMin	35.1	2.0	1.7	1.1	0.9	0.4	1.5	1.3	1.5	0.4	0.1	0.7	0.9	(17)
	PrecStd	6.5	1.6	1.4	2.2	1.5	2.0	2.1	2.0	2.2	2.6	2.5	1.9	1.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.9	78.7	84.0	88.4	95.1	99.1	99.3	99.2	95.5	89.4	79.7	76.6	(19)
		MCWB	64.1	64.9	64.1	66.4	70.3	72.3	74.3	72.8	73.2	73.9	66.1	67.7	(20)
	2%	DB	70.1	73.2	80.8	85.1	90.9	95.9	96.7	94.8	92.1	84.7	75.3	72.7	(21)
		MCWB	62.1	60.5	63.2	65.9	70.5	74.4	75.9	74.7	73.3	70.0	64.4	66.0	(22)
	5%	DB	66.0	69.8	76.1	81.8	88.0	92.9	93.5	92.5	90.1	81.6	72.4	68.5	(23)
		MCWB	60.3	58.7	60.8	63.8	70.2	73.8	75.5	76.1	72.8	69.4	62.5	64.3	(24)
	10%	DB	62.7	65.0	72.3	79.0	84.3	90.3	91.2	90.1	87.3	78.7	68.6	64.5	(25)
		MCWB	57.0	56.8	59.6	62.9	69.4	73.7	75.8	75.2	72.7	67.2	61.6	60.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.2	67.6	69.3	72.8	76.5	80.2	81.8	81.9	81.0	77.7	72.6	72.0	(27)
		MCDB	71.9	74.1	77.5	81.1	85.6	91.2	91.9	90.2	86.3	81.2	75.0	74.0	(28)
	2%	WB	64.1	65.0	66.6	70.3	74.5	78.7	80.3	79.8	78.1	74.7	68.9	68.1	(29)
		MCDB	67.7	70.8	74.0	77.7	83.8	89.2	89.1	87.8	85.3	80.3	72.1	69.9	(30)
	5%	WB	61.6	62.4	64.5	68.1	72.9	77.5	79.1	78.4	76.3	73.0	66.0	65.0	(31)
		MCDB	64.7	66.9	72.0	75.5	82.0	87.6	87.3	86.2	83.6	78.1	69.8	67.9	(32)
	10%	WB	58.2	59.5	62.3	66.2	71.3	76.3	78.0	77.4	74.9	70.5	62.7	61.4	(33)
		MCDB	61.9	63.4	69.5	73.8	79.7	85.3	86.5	85.2	82.3	75.6	67.0	64.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.4	22.2	24.3	24.9	23.1	21.4	20.1	19.2	21.3	23.5	24.4	21.0	(35)
		MCDBR	23.8	25.4	27.5	27.5	26.7	25.3	23.3	22.8	25.2	25.4	27.9	23.4	(36)
	5% WB	MCWBR	16.3	14.9	13.2	11.0	9.9	7.8	7.4	7.7	8.8	12.8	16.8	16.6	(37)
		MCWBR	19.3	19.8	22.1	21.2	21.4	20.9	19.7	19.1	20.0	18.9	20.1	19.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.318	0.325	0.360	0.393	0.455	0.489	0.532	0.525	0.429	0.390	0.345	0.315	(40)
		taud	2.513	2.490	2.421	2.313	2.184	2.136	2.036	2.042	2.296	2.379	2.460	2.551	(41)
	Ebn at Noon		285	294	291	284	266	256	244	244	265	267	271	280	(42)
		Edh at Noon	27	30	35	41	47	49	54	53	39	34	28	24	(43)
(44) All-Sky Solar Radiation	RadAvg		887	1078	1426	1784	1961	2016	1918	1754	1528	1285	974	764	(44)
	RadStd		49	120	103	121	159	148	135	81	135	151	81	77	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+331	+234

Nomenclature: See separate page